

Math Camp 2026 — Day 1 Exercise

Ten short questions

About 30–40 minutes.

Part A on paper. Part B in `day1-exercise.R`, saved in your project folder.

Part A — Algebra and functions

1. Summation and product notation

(a) Evaluate $\sum_{i=1}^4 (2i - 1)$.

(b) Evaluate $\prod_{i=1}^3 2i$.

2. Exponents and logarithms

(a) Simplify: $x^5 \cdot x^{-2}$

(b) Solve for x : $2^x = 32$

(c) Simplify to a single term: $\log(ab) - \log(b)$

3. Functions and domains

(a) Give the domain of $f(x) = \sqrt{x - 2}$.

(b) Give the domain of $g(x) = \ln(5 - x)$.

(c) With $f(x) = x^2$ and $g(x) = x + 3$, compute $f(g(1))$.

4. Log-odds

(a) A candidate wins with probability $p = 0.8$. Compute the odds $\frac{p}{1-p}$, then the log-odds $\ln \frac{p}{1-p}$.

(b) If the log-odds equal 0, what is p ?

5. Limits

(a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Hint: factor the numerator. $x^2 - 9 = (x - 3)(x + 3)$.

(b) Is that function defined **at** $x = 3$? (Just yes or no.)

Part B — R

Open `day1-exercise.R` and write your answers there.

6. Vectors and indexing

Create `v <- c(2, 7, 1, 8, 2, 8)`, then:

- (a) How many elements does it have?
- (b) Return the second element.
- (c) Run `v[-1]`. What comes back, and what does the minus sign do?

7. Vectorized arithmetic

Still with `v`:

- (a) Multiply every element by 3, in one line.
- (b) Produce the `TRUE/FALSE` vector for “is greater than 5”.
- (c) Return the elements that are greater than 5.

8. Types and factors

- (a) Use `class()` on 42, “hello”, and `TRUE`.
- (b) Run these three lines. Which one gives you the actual numbers?

```
x <- factor(c("10", "20", "30"))
as.numeric(x)
as.numeric(as.character(x))
```

9. Data frames

- (a) Build a data frame `df` with three countries: a `country` column, a `region` column, and a numeric `exports` column. Make up the numbers.
- (b) Run `str(df)`.
- (c) Return only the rows from one region.

10. Writing a function

- (a) Write a function `logit(p)` that returns $\ln \frac{p}{1-p}$.
- (b) Run `logit(0.8)`. Does it match your answer to question 4(a)?
- (c) Each of these two lines is broken. Fix them.

```
Mean(v)
mean(vv)
```

If you want one more

Question 1(a) in R, in a single line. *Hint: start from 1:4.*